

Explainer: Problem 6 (Rudin 7.4) — Analyzing $\sum \frac{1}{1+n^2x}$

The series

$$f(x) = \sum_{n=1}^{\infty} \frac{1}{1+n^2x}.$$

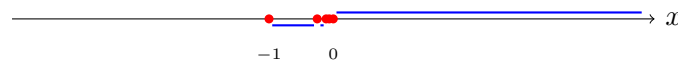
You need to answer five questions:

1. For what x does the series converge absolutely?
2. Where does it converge uniformly?
3. Where does it fail to converge uniformly?
4. Is f continuous where the series converges?
5. Is f bounded?

First: where can it even converge?

Look at the n th term: $\frac{1}{1+n^2x}$.

- $x > 0$: Term behaves like $1/(n^2x)$, so compare with $\sum 1/n^2$. **Converges absolutely.**
- $x = 0$: Each term is 1. $\sum 1$ diverges.
- $x < 0$: Then $1 + n^2x = 0$ exactly when $n^2 = -1/x$. So if $-1/x$ is a perfect square, a term is undefined. Otherwise the terms grow like $-1/(n^2|x|)$ and the sum converges. But you must *exclude* the reciprocals of negative squares: $x \neq -1/n^2$ for any n .



Red dots: bad points ($x = 0$ and $x = -1/n^2$).

Uniform convergence

Use the **Weierstrass M -test**: if $|\text{term}_n(x)| \leq M_n$ with $\sum M_n < \infty$, then uniform convergence.

On $[\delta, \infty)$ for any $\delta > 0$: bound $|\frac{1}{1+n^2x}| \leq \frac{1}{n^2\delta}$, M -test wins. **Uniform.**

On $(0, \infty)$ (not bounded away from 0): as $x \rightarrow 0^+$, each term $\rightarrow 1$, and any tail $\sum_{n=N}^{\infty}$ can be forced large by choosing x small. **Not uniform.**

Near negative roots $-1/n^2$: the individual term $1/(1+n^2x)$ blows up, so no uniform convergence on any interval containing such a point.

On any interval bounded away from 0 and from every $-1/n^2$: similar M -test argument works.

Continuity

Uniform limit of continuous functions is continuous (Theorem 7.12). The series converges uniformly on every compact set avoiding the bad points — continuity is a *local* property, so f is continuous on its domain.

Boundedness

Check behavior as $x \rightarrow 0^+$: each term $\rightarrow 1$. Partial sums $\rightarrow N$ at $x = 0^+$. In fact $f(x) \rightarrow +\infty$ as $x \rightarrow 0^+$. **So f is not bounded.**

Strategy summary

1. Pin down the domain (exclude $x = 0, -1/n^2$).
2. Use M -test on intervals bounded away from the bad points.
3. Show non-uniformity by exhibiting x -values where tails stay $\geq$ some fixed constant.
4. Continuity from uniform convergence on compact pieces of the domain.
5. Boundedness: analyze $x \rightarrow 0^+$.